

A Survey of Privacy-Preserving Machine Learning with Federated Learning and Differential Privacy

InteractiveSurvey

Abstract

The rapid proliferation of distributed data sources has established Federated Learning as a pivotal paradigm for collaborative model training, yet shared gradients remain susceptible to sophisticated inference attacks that compromise sensitive information. Consequently, ensuring rigorous privacy guarantees necessitates the integration of formal frameworks capable of withstanding adversarial scrutiny while maintaining model utility. This survey provides a comprehensive examination of Privacy-Preserving Machine Learning, specifically focusing on the synergistic integration of Federated Learning and Differential Privacy to bridge the gap between theoretical bounds and implementation realities. We systematically analyze how noise injection, sensitivity control, and cryptographic protocols can be optimized to balance stringent mathematical guarantees with the practical necessities of convergence speed in heterogeneous networks. The review establishes a unified taxonomy of mechanisms, critically evaluating adaptive clipping strategies, secure aggregation via cryptographic masking, and privacy amplification through subsampling and shuffling. Furthermore, it synthesizes insights on leveraging quantization and sparsification to inherently enhance privacy while alleviating communication bottlenecks, alongside robust defenses against gradient leakage and data poisoning. By integrating mechanism design with adversarial robustness, this work constructs a holistic framework for understanding the interplay between privacy preservation and model integrity. Ultimately, this survey offers a comprehensive roadmap for developing resilient federated systems that can withstand both passive inference attacks and active poisoning attempts, guiding future innovations in secure collaborative intelligence.

1 Introduction

The rapid proliferation of distributed data sources has catalyzed the adoption of Federated Learning (FL) as a paradigm for collaborative model training without centralizing raw data [1]. By enabling multiple clients to train models locally and share only parameter updates, FL ostensibly mitigates traditional privacy risks associated with data aggregation [2]. However, emerging research demonstrates that shared gradients remain vulnerable to sophisticated inference attacks, including model inversion and membership inference, which can reconstruct sensitive training data from seemingly innocuous updates. Consequently, ensuring rigorous privacy guarantees in distributed environments requires more than mere data decentralization; it demands the integration of formal privacy frameworks capable of withstanding adversarial scrutiny while maintaining model utility.

This survey paper provides a comprehensive examination of Privacy-Preserving Machine Learning, specifically focusing on the synergistic integration of Federated Learning and Differential Privacy (DP) [3]. While existing literature often treats these domains in isolation or addresses them superficially, this work delves into the intricate mechanisms required to deploy DP within FL architectures effectively. The primary objective is to analyze how noise injection, sensitivity control, and cryptographic protocols can be optimized to balance the stringent mathematical guarantees of differential privacy with the practical necessities of convergence speed and accuracy in heterogeneous network settings [4]. By bridging the gap between theoretical privacy bounds and implementation realities, this review aims to establish a unified understanding of current state-of-the-art methodologies.

The initial portion of this review scrutinizes the core mechanisms of differential privacy, empha-

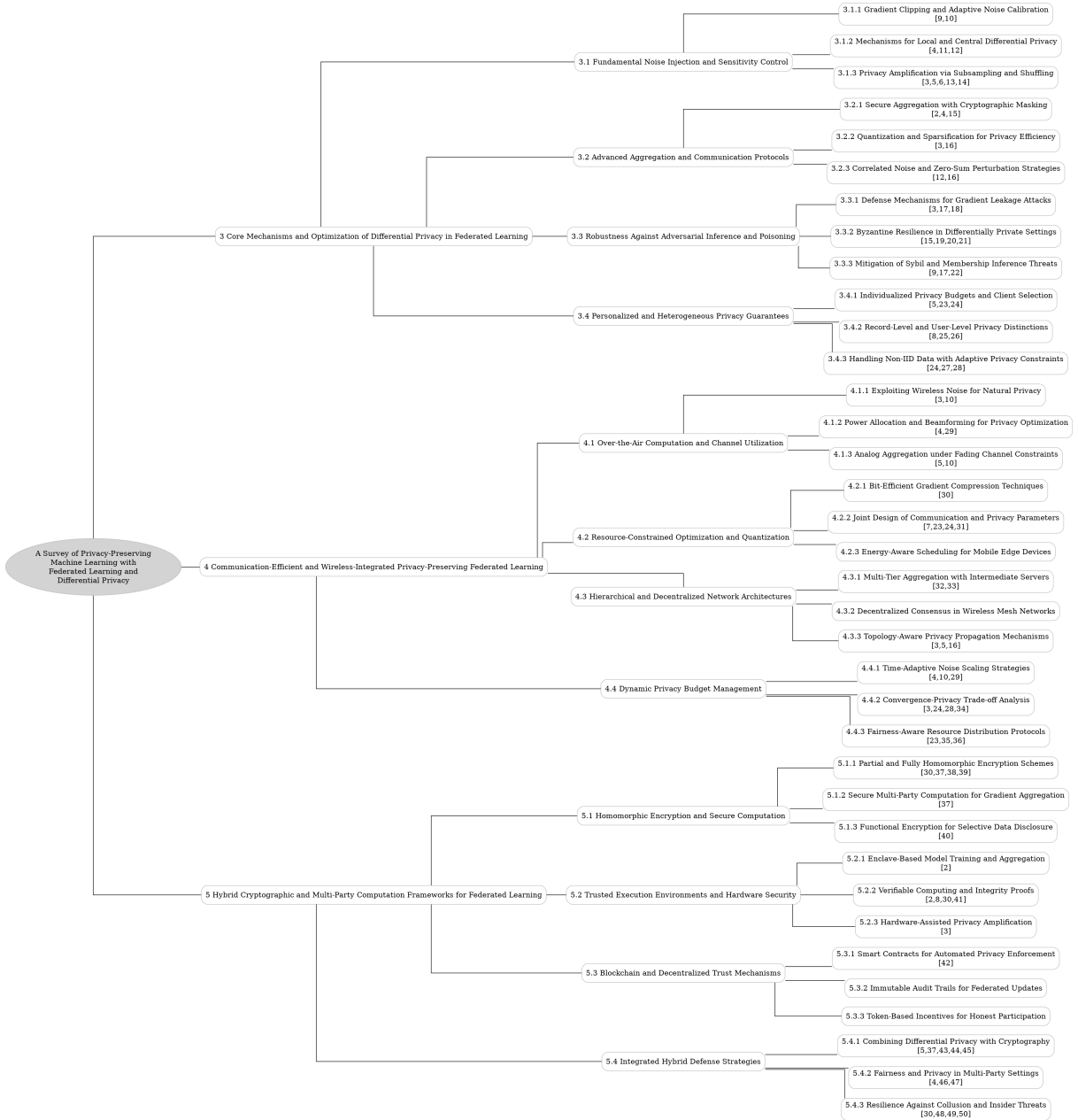


Figure 1: The outline of our survey: A Survey of Privacy-Preserving Machine Learning with Federated Learning and Differential Privacy

$$\frac{\Pr[M(D) = O]}{\Pr[M(D') = O]} \leq e^\epsilon$$

Figure 2: Chart from 'Privacy-Preserving Large Language Models: Mechanisms, Applications, and Future Directions' (arXiv:2412.06113)

$$\sigma_D = \begin{cases} \frac{2cC\sqrt{T^2-L^2N}}{mN\epsilon} & T > L\sqrt{N} \\ 0 & T \leq L\sqrt{N} \end{cases}$$

Figure 3: Chart from 'Adaptive Differential Privacy in Federated Learning: A Priority-Based Approach' (arXiv:2401.02453)

sizing fundamental noise injection and sensitivity control strategies [5]. It explores the critical trade-offs inherent in gradient clipping, where static thresholds often lead to either biased convergence or excessive noise, and contrasts these with adaptive calibration techniques that dynamically adjust to gradient distributions. Furthermore, the discussion differentiates between Local and Central Differential Privacy models, evaluating their respective trust assumptions and utility profiles [6]. Special attention is given to privacy amplification techniques, such as subsampling and shuffling, which leverage probabilistic exclusion and anonymization to tighten privacy bounds without proportionally increasing noise variance, thereby enhancing the overall efficiency of the learning process.

Subsequent analysis shifts toward advanced aggregation and communication protocols designed to harmonize privacy with computational and bandwidth constraints. This section details secure aggregation methods utilizing cryptographic masking to hide individual contributions through secret sharing, addressing the challenges of integrating continuous noise distributions with dis-

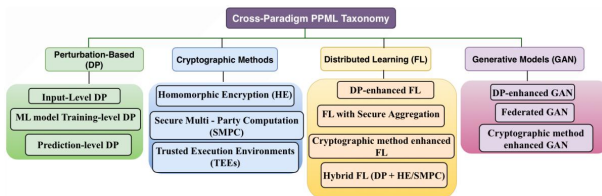


Figure 4: Chart from 'Privacy-Preserving Machine Learning for IoT: A Cross-Paradigm Survey and Future Roadmap' (arXiv:2603.13570)

$$w_{t+1} = w_t + \frac{1}{m_t} \sum_{i=0}^{m_t-1} \frac{\Delta w_t}{\max(1, \frac{\|\Delta w_i\|_2}{S}, \frac{\|\Delta w_i\|_2}{M})} + N(0, \sigma^2 S^2)$$

Figure 5: Chart from 'Fair Differentially Private Federated Learning Framework' (arXiv:2305.13878)

$$W_{t+1} = \sum_{i=1}^K \alpha_i W_t^i$$

Figure 6: Chart from 'Privacy Preserved Federated Learning with Attention-Based Aggregation for Biometric Recognition' (arXiv:2510.01113)

crete arithmetic. Concurrently, the review investigates the dual role of quantization and sparsification, illustrating how compression techniques can inherently amplify privacy by reducing sensitivity while alleviating communication bottlenecks. The discourse extends to correlated noise and zero-sum perturbation strategies, which introduce structured dependencies among clients to minimize cumulative noise magnitude, offering superior utility-privacy trade-offs compared to traditional independent perturbation methods in high-dimensional settings.

Finally, the survey addresses the critical dimension of robustness against adversarial threats, specifically focusing on defense mechanisms for gradient leakage and data poisoning attacks [7]. It evaluates how differential privacy serves as a foundational barrier against inference attacks that attempt to reconstruct private data from shared updates, while also examining its efficacy in mitigating the impact of malicious clients injecting poisoned gradients. By synthesizing insights from mechanism design, communication efficiency, and adversarial robustness, this work constructs a holistic framework for understanding the interplay between privacy preservation and model integrity. The review concludes by identifying persistent challenges and outlining promising directions for future research in scalable, robust, and privacy-preserving distributed intelligence.

The primary contributions of this survey are threefold. First, it offers a systematic taxonomy of differential privacy mechanisms tailored specifically for federated environments, categorizing ap-

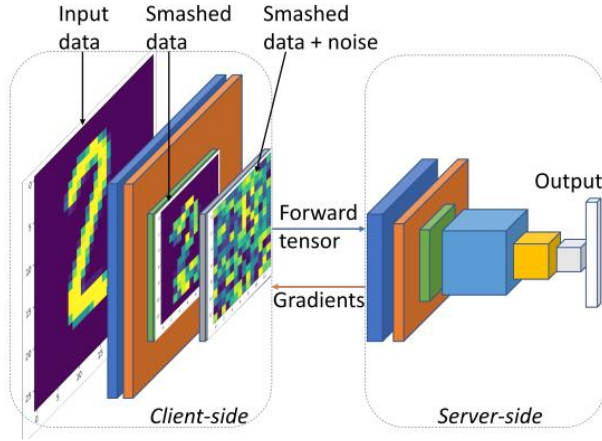


Figure 7: Chart from 'Enhancing Accuracy-Privacy Trade-off in Differentially Private Split Learning' (arXiv:2310.14434)

proaches by their trust models, noise structures, and amplification techniques [5]. Second, it provides a critical comparative analysis of recent advancements in adaptive clipping, secure aggregation, and compression-aware privacy, highlighting the theoretical underpinnings and practical limitations of each method. Third, by integrating discussions on adversarial robustness with core privacy mechanisms, this paper establishes a comprehensive roadmap for developing resilient federated systems that can withstand both passive inference attacks and active poisoning attempts, thereby guiding future innovations in secure collaborative machine learning [8].

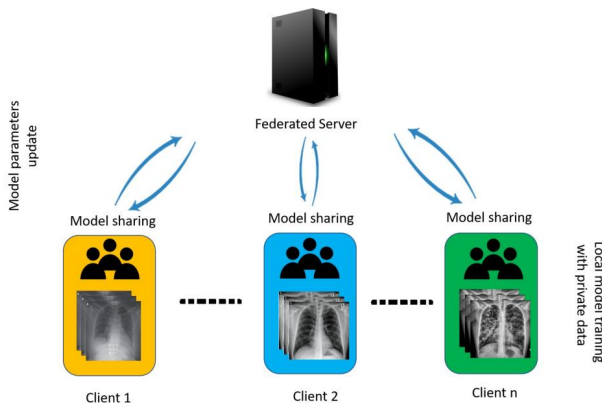


Figure 8: Chart from 'Differential Privacy by Design Framework for Federated Machine Learning Systems: A Case Study on COVID-19 Imaging Data' (arXiv:2010.06177)

2 Core Mechanisms and Optimization of Differential Privacy in Federated Learning

2.1 Fundamental Noise Injection and Sensitivity Control

2.1.1 Gradient Clipping and Adaptive Noise Calibration

Gradient clipping serves as the foundational mechanism for bounding sensitivity in differentially private stochastic gradient descent, ensuring that the addition of Gaussian noise yields rigorous privacy guarantees [9]. By constraining the ℓ_2 norm of individual gradients to a fixed threshold C , the system prevents any single data point from exerting disproportionate influence on the model update. However, the selection of this clipping bound presents a critical trade-off: an underestimated threshold introduces significant gradient bias, potentially causing model divergence or convergence to suboptimal minima, while an overestimated threshold necessitates excessive noise injection to maintain the same privacy budget, thereby degrading utility.

To mitigate these static limitations, recent advancements prioritize adaptive noise calibration and dynamic threshold estimation. Instead of relying on fixed hyperparameters, adaptive strategies analyze gradient distributions across training rounds to identify optimal clipping norms, often utilizing percentiles of observed gradient magnitudes to account for heterogeneity among clients in federated settings. These methods frequently integrate with moments accountant techniques to track tighter privacy loss bounds, allowing for the progressive reduction of noise scales as gradients converge. Such dynamic allocation ensures that the noise variance is proportional to the actual sensitivity of the current training state rather than a worst-case theoretical bound.

Furthermore, novel approaches explore modifying the sequence of operations or the objective function itself to naturally constrain sensitivity without aggressive clipping. Techniques such as first-order approximations with inherent norm constraints and adaptive noise coefficient decay facilitate smoother convergence while preserving differential privacy. By concurrently optimizing local data transformations or employing discrete Gaussian mechanisms with compression-aware sensitivity analysis, these methods address the com-

pounding effects of clipping bias and noise variance. Ultimately, the synergy between precise clipping bound estimation and adaptive noise calibration is essential for balancing the stringent requirements of privacy preservation with the practical necessities of model accuracy and convergence speed [10].

2.1.2 Mechanisms for Local and Central Differential Privacy

Local Differential Privacy (LDP) operates under a stringent trust model where each client perturbs their data locally before transmission, ensuring the central server never accesses raw information [11]. Mechanisms such as the randomized response and Laplace noise injection are applied directly to individual gradients or model parameters, providing robust protection against honest-but-curious aggregators. While LDP eliminates the need for a trusted third party, it often necessitates injecting significant noise to maintain privacy guarantees, which can severely degrade model utility and slow convergence in federated learning scenarios, particularly when the number of participants is limited.

$$\epsilon \geq \frac{\sqrt{2 \ln(1.25/\delta)}}{\frac{\sigma}{\Delta_2 f}}, \quad \Delta_2(f) = \max_{d, d' \in \mathcal{D}} \|f(d) - f(d')\|_2$$

Figure 9: Chart from 'Toward the Tradeoffs between Privacy, Fairness and Utility in Federated Learning' (arXiv:2311.18190)

In contrast, Central Differential Privacy (CDP) assumes a trusted aggregation server that collects unperturbed updates from clients and applies noise only to the aggregated global model or intermediate sums [4]. This approach leverages the properties of secure aggregation to amplify privacy, allowing for substantially less noise injection per client compared to LDP while achieving equivalent overall privacy bounds [12]. By deferring perturbation until the aggregation phase, CDP mechanisms preserve higher model accuracy and faster convergence rates, making them preferable in settings where the central coordinator is verified and secure communication channels are established.

Recent advancements have explored hybrid frameworks, such as the shuffle model, which bridges the gap between local and central paradigms by introducing an anonymous shuffler between clients and the server. This architecture enables privacy amplification through anonymity,

$$SecAgg(g_i^{clip} + z_i, \mathcal{U}_t) = (\sum_i g_i) + z,$$

Figure 10: Chart from 'Communication Efficient, Differentially Private Distributed Optimization using Correlation-Aware Sketching' (arXiv:2507.03545)

offering central-level utility with weaker trust assumptions than pure CDP. Furthermore, sophisticated accounting methods like Rényi Differential Privacy are increasingly employed to tightly track privacy loss across multiple composition rounds, optimizing the trade-off between the rigorous indistinguishability guarantees of LDP and the superior utility characteristics of central approaches.

2.1.3 Privacy Amplification via Subsampling and Shuffling

Privacy amplification via subsampling and shuffling represents a pivotal advancement in differentially private federated learning, enabling stronger privacy guarantees with reduced noise injection [3]. Subsampling leverages the probabilistic exclusion of data points or clients from the training process, effectively scaling down the privacy loss parameters by a factor dependent on the sampling rate. This mechanism relies on the principle that an adversary cannot determine whether a specific individual's data was included in the computed aggregate, thereby tightening the bounds on privacy leakage compared to mechanisms operating on the full dataset.

Complementing subsampling, the shuffling protocol introduces an anonymization layer where locally perturbed updates are randomly permuted by a trusted intermediary before reaching the untrusted aggregator [13]. This process breaks the direct link between specific model updates and their originating clients, transforming local differential privacy guarantees into significantly stronger central differential privacy bounds [5]. Theoretical analyses demonstrate that shuffling provides substantial amplification, particularly in high-dimensional settings, by exploiting the anonymity set created among the participating clients to mask individual contributions more effectively than noise alone [14].

Recent research focuses on the composability of these two techniques to maximize the privacy-utility trade-off [6]. While both methods independently offer amplification, their joint applica-

$$[z] = \sum_{i=1}^K [w_i] [b'_i].$$

Figure 11: Chart from 'RAIN: Secure and Robust Aggregation under Shuffle Model of Differential Privacy' (arXiv:2603.03108)

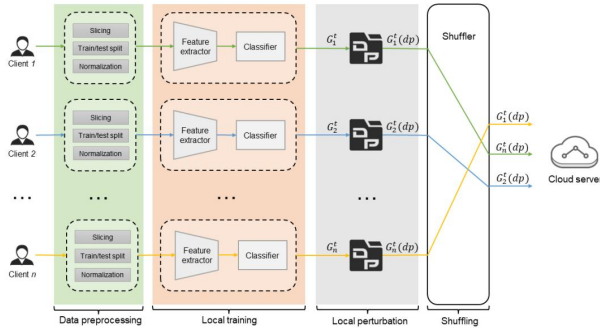


Figure 12: Chart from 'Shuffled Differentially Private Federated Learning for Time Series Data Analytics' (arXiv:2307.16196)

tion requires rigorous accounting, often utilizing Rényi differential privacy to derive tight bounds on cumulative privacy loss over multiple training rounds. Challenges remain in ensuring these amplification effects hold for multi-dimensional vector updates without diminishing returns as dimensionality increases. Consequently, novel protocols incorporating dummy padding and advanced composition theorems are being developed to bridge these mechanisms, ensuring robust privacy preservation while maintaining model convergence and accuracy in distributed environments.

2.2 Advanced Aggregation and Communication Protocols

2.2.1 Secure Aggregation with Cryptographic Masking

Secure aggregation with cryptographic masking eliminates the need for a trusted central server by leveraging secure multi-party computation protocols to compute the sum of client updates without revealing individual contributions [2]. The foundational approach employs pairwise random masks generated via secret sharing, where each client adds a unique mask to their local update be-

fore transmission. These masks are constructed such that they cancel out perfectly during the aggregation phase at the server, ensuring that only the global sum is revealed while individual gradients remain information-theoretically hidden from both the aggregator and colluding clients.

A critical technical challenge in this domain arises from the incompatibility between continuous differential privacy noise distributions and the discrete arithmetic required by cryptographic protocols [4]. Since secure aggregation typically operates over finite fields to manage computational complexity and support modular arithmetic, directly applying continuous Gaussian noise is infeasible. To resolve this, recent methodologies project continuous values into quotient groups or utilize discrete Gaussian mechanisms, allowing the integration of rigorous differential privacy guarantees within the cryptographic masking framework without compromising the cancellation properties essential for correct model convergence.

Furthermore, these protocols are designed to be robust against dynamic network conditions, specifically addressing the issue of random client dropouts common in cross-device federated learning. Advanced masking schemes incorporate recovery mechanisms that allow the server to reconstruct the necessary mask secrets for surviving clients while neutralizing the masks of dropped participants, ensuring the aggregation remains unbiased. By combining these cryptographic guarantees with differential privacy, the system achieves a strong privacy-utility trade-off, protecting against gradient leakage and inference attacks even under worst-case assumptions regarding server honesty and collusion among subsets of clients [15].

$$\sigma_i \geq c_2 \frac{q_i \sqrt{T \ln(1/\delta_i)}}{\epsilon_i}.$$

Figure 13: Chart from 'Federated Learning Meets Fairness and Differential Privacy' (arXiv:2108.09932)

2.2.2 Quantization and Sparsification for Privacy Efficiency

Quantization and sparsification serve as dual-purpose mechanisms in federated learning, si-

multaneously addressing communication bottlenecks and enhancing differential privacy guarantees [3]. While traditional approaches often treat compression and privacy as independent objectives, recent advancements demonstrate that per-parameter quantization and gradient sparsification can inherently amplify privacy through subsampling effects. By transmitting only a subset of significant gradients or reduced-precision values, these techniques effectively reduce the sensitivity of the aggregated updates, thereby allowing for the injection of less noise to satisfy a fixed privacy budget compared to full-precision, dense transmission protocols.

However, naive application of these compression strategies introduces complex trade-offs between utility, communication cost, and privacy accounting. Standard per-parameter quantization schemes often incur significant overhead when attempting to maintain optimal accuracy under centralized differential privacy constraints. To mitigate this, linear schemes based on sparse random projections have been proposed to fundamentally lower communication costs while preserving model convergence. Furthermore, advanced protocols leverage gradient sparsification to enable privacy amplification by subsampling, where perturbing only a fraction of dimensions allows each selected component to utilize a larger portion of the privacy budget, significantly tightening the overall privacy loss bounds via Rényi differential privacy analysis [16].

$$y_{i,j} \leftarrow \begin{cases} \mathcal{R}_{\epsilon_{lk}}(\tilde{x}_{i,j}), & j \in S_{top} \\ \omega_{\mathcal{R}}, & j \in S_{non} \end{cases}$$

Figure 14: Chart from 'FLAME: Differentially Private Federated Learning in the Shuffle Model' (arXiv:2009.08063)

The integration of these methods requires rigorous theoretical frameworks to ensure that compression-induced distortions do not compromise the formal privacy guarantees. Recent literature emphasizes the necessity of holistic designs where privacy-aware compression mechanisms are optimized for specific communication budgets rather than simply layering standard compression atop existing privacy mechanisms. By carefully calibrating clipping norms and noise multipliers in conjunction with sparsification ra-

tios, it is possible to achieve superior utility-privacy trade-offs. This approach not only minimizes the variance introduced by noise but also ensures that the resulting models remain robust against inference attacks, even in scenarios involving non-i.i.d. data distributions and stringent privacy requirements.

2.2.3 Correlated Noise and Zero-Sum Perturbation Strategies

Correlated noise strategies fundamentally diverge from traditional independent perturbation by introducing structured dependencies among clients to optimize the privacy-utility tradeoff. Unlike standard mechanisms where noise accumulates linearly, correlated approaches, such as tree aggregation techniques, inject noise vectors that are statistically dependent across iterations or participants. This methodology allows for tighter tracking of privacy loss through advanced composition theorems, effectively reducing the total noise magnitude required to satisfy a global differential privacy budget. By carefully dividing the privacy budget and correlating noise terms, these strategies mitigate the severe utility degradation often observed in high-dimensional federated learning settings where independent Gaussian noise would otherwise overwhelm the signal [12].

Zero-sum perturbation represents a specialized subclass of correlated noise designed explicitly for secure aggregation protocols. In this framework, clients generate pairwise noise terms that sum to zero across the entire population, ensuring that the aggregate model update remains unbiased while individual contributions remain obfuscated. Typically, each client adds individual Gaussian noise alongside $N - 1$ pairwise terms constructed such that they cancel out upon summation at the server. This architecture prevents an untrusted aggregator from isolating specific updates, provided that straggler effects are managed, as the absence of participating clients could otherwise allow the server to mathematically resolve and cancel the pairwise components, potentially exposing raw gradients.

The theoretical foundation of these strategies often leverages discrete mechanisms like the Skellam distribution, which arises from the difference of two independent Poisson variables and maintains closure under summation. This property is critical for federated environments, as it ensures that the sum of discrete noisy updates follows a

known distribution without requiring modular clipping that introduces bias. Furthermore, zero-sum and correlated approaches facilitate privacy amplification by shuffling or aggregation, yielding sharper Renyi Differential Privacy bounds compared to independent noise addition [16]. Consequently, these methods enable the use of smaller noise scales for equivalent privacy guarantees, significantly improving model convergence rates and final accuracy in distributed learning scenarios.

2.3 Robustness Against Adversarial Inference and Poisoning

2.3.1 Defense Mechanisms for Gradient Leakage Attacks

Gradient leakage attacks pose a severe threat to federated learning by enabling adversaries to reconstruct private training data from shared model updates [3]. To mitigate this, Differential Privacy (DP) has emerged as a foundational defense, primarily through gradient perturbation mechanisms [17]. By injecting calibrated noise, such as Gaussian or Laplacian distributions, into local gradients before transmission, these methods mathematically bound the information leakage regarding any individual sample. However, effective implementation requires careful gradient clipping to control sensitivity and prevent exploding gradients, ensuring that the added noise provides rigorous privacy guarantees without completely obliterating model utility.

$$C_t = G_t \times G_p,$$

Figure 15: Chart from 'Securing Health Data on the Blockchain: A Differential Privacy and Federated Learning Framework' (arXiv:2405.11580)

Beyond standalone noise injection, advanced frameworks integrate DP with Secure Multiparty Computation (SMPC) to create robust, malicious-secure aggregation protocols. In these hybrid approaches, cryptographic masking ensures that the server only accesses the aggregate sum of gradients, while individual contributions remain confidential even from the aggregator. This combination significantly reduces single-point data leakage risks and thwarts reconstruction attempts by both honest-but-curious servers and colluding clients. Furthermore, recent innovations employ adaptive

noise scales and shuffler-based architectures to enhance the privacy-utility trade-off, allowing for stronger protection against deep leakage from gradients while maintaining convergence rates comparable to non-private training [18].

$$w(t+1) = w(t) - \eta \sum_{k=1}^{K_t} \frac{n_k}{n} \nabla w_k(t),$$

Figure 16: Chart from 'Data Poisoning and Leakage Analysis in Federated Learning' (arXiv:2409.13004)

Comprehensive defense strategies also address complex scenarios involving active attackers who may alter updates to amplify leakage or induce poisoning. Techniques such as combining local DP with Byzantine fault tolerance leverage random noise to obscure malicious patterns, effectively neutralizing attacks that exploit gradient anomalies. Additionally, methods extending protection to both feature and label holders in vertical federated learning ensure end-to-end confidentiality during forward and backward propagation. These multi-layered defenses collectively establish a resilient infrastructure, preventing data reconstruction and ensuring that the trained models remain insensitive to the participation of any specific client, thereby upholding strict privacy standards in distributed environments.

2.3.2 Byzantine Resilience in Differentially Private Settings

Integrating Byzantine fault tolerance with differential privacy introduces complex trade-offs between robustness, utility, and privacy guarantees. While differential privacy inherently obscures individual contributions through calibrated noise, this stochasticity can inadvertently mask malicious gradients injected by Byzantine actors, complicating detection mechanisms [19]. Traditional robust aggregation rules, which often rely on identifying statistical outliers, may fail when privacy noise increases the variance of legitimate updates, thereby reducing the distinguishability between honest and poisoned models. Consequently, naive combinations of these defenses often result in significant degradation of model convergence rates or require impractically large privacy budgets to maintain utility.

Recent advancements address these challenges by leveraging the properties of differential privacy to enhance robustness rather than viewing it solely

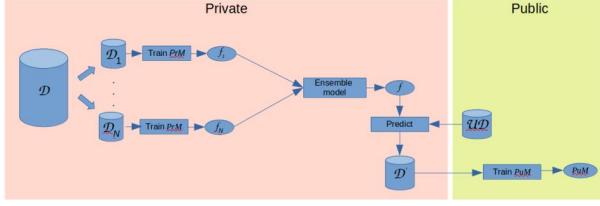


Figure 17: Chart from 'Differential Privacy: What is all the noise about?' (arXiv:2205.09453)

as an impediment [20]. Specific approaches combine local differentially private stochastic gradient descent with Byzantine-resilient aggregation protocols, utilizing the random noise inherent in privacy mechanisms to dilute the impact of coordinated attacks. By carefully tuning the noise multiplier and gradient clipping thresholds, these methods construct aggregation rules that effectively thwart reconstruction and model poisoning attacks while preserving the theoretical bounds of privacy. Furthermore, the adoption of Rényi differential privacy has enabled tighter composition analysis, allowing for more precise accounting of cumulative privacy loss across multiple rounds of asynchronous decentralized training without compromising security against adaptive adversaries [21].

$$\mathbf{g}'_k^t = \frac{1}{t\eta} (\theta^t - \theta_k^t)$$

Figure 18: Chart from 'User-Level Differential Privacy against Attribute Inference Attack of Speech Emotion Recognition in Federated Learning' (arXiv:2204.02500)

$$+ \frac{d\eta L \sigma^2}{pqnc} \frac{1}{T} \sum_{t=0}^{T-1} \frac{1}{(\gamma^{(t)})^2} + \sqrt{\frac{d\eta L}{pqn} \frac{1}{T} \sum_{t=0}^{T-1} \frac{\sigma}{\gamma^{(t)}}}.$$

Figure 19: Chart from 'WHEN DIFFERENTIAL PRIVACY MEETS WIRELESS FEDERATED LEARNING: AN IMPROVED ANALYSIS FOR PRIVACY AND CONVERGENCE' (arXiv:2603.19040)

Despite these progressions, achieving optimal efficiency in heterogeneous environments remains an open problem, particularly when balancing strict privacy constraints with the need for high-dimensional gradient verification. Current literature suggests that hybrid frameworks, which integrate secure multi-party computation with differ-

ential privacy, offer a promising direction for mitigating the limitations of purely noise-based defenses [15]. However, the computational overhead associated with verifying encrypted updates under strict privacy budgets necessitates further optimization. Future research must focus on developing adaptive mechanisms that dynamically adjust privacy parameters based on detected threat levels, ensuring resilient convergence even in scenarios with high fractions of malicious participants and non-independent and identically distributed data distributions.

2.3.3 Mitigation of Sybil and Membership Inference Threats

Membership inference attacks exploit statistical discrepancies in model outputs to determine if a specific sample contributed to the training set, posing severe risks in federated learning where adversaries may include malicious clients or the central server [17]. These threats are often compounded by Sybil attacks, wherein an adversary fabricates numerous fake identities to dominate the aggregation process, thereby amplifying inference capabilities or manipulating global model convergence. The synergy between these vectors allows attackers to bypass standard defenses, as a sufficient number of Sybil nodes can overwhelm honest participants, rendering traditional anomaly detection ineffective and facilitating high-success rate privacy breaches.

To mitigate these dual threats, Differential Privacy (DP) has emerged as a foundational defense mechanism, specifically through the injection of calibrated noise into gradients or model updates prior to aggregation [9]. Techniques such as DP-Stochastic Gradient Descent (DP-SGD) and the distributed discrete Gaussian mechanism ensure that the contribution of any single client remains indistinguishable, theoretically bounding the advantage an adversary gains from membership inference. Furthermore, privacy amplification by shuffling or secure aggregation protocols leverages the noise introduced by honest clients to enhance overall privacy guarantees, provided the cohort of legitimate participants remains sufficiently large to dilute the influence of Sybil entities.

However, relying solely on client-side noise injection is insufficient against powerful adversaries controlling a significant fraction of the network. Advanced mitigation strategies increasingly integrate cryptographic primitives like Secure Multi-

Party Computation (SMPC) and Homomorphic Encryption with adaptive DP to enforce strict access controls and verify client authenticity without exposing raw data [22]. Recent frameworks propose hybrid approaches that dynamically adjust noise scales based on trust scores or employ zero-knowledge proofs to validate update integrity, ensuring robustness even when the server is semi-honest. Despite these advancements, maintaining the delicate balance between rigorous privacy protection, resistance to Sybil domination, and model utility remains a critical open challenge in decentralized learning environments.

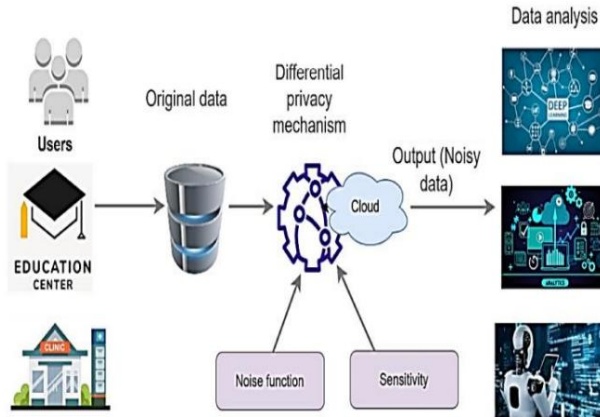


Figure 20: Chart from 'Privacy-Preserving Data Processing in Cloud : From Homomorphic Encryption to Federated Analytics' (arXiv:2601.06710)

2.4 Personalized and Heterogeneous Privacy Guarantees

2.4.1 Individualized Privacy Budgets and Client Selection

Traditional federated learning frameworks often rely on uniform sampling assumptions and global privacy budgets, which fail to address the dynamic nature of cross-device environments where client dropout is prevalent [23]. Recent advancements have shifted towards individualized privacy budgets, allowing heterogeneous ϵ_k -LDP settings for each participant based on specific re-identification risks or data sensitivity. This personalized approach necessitates robust theoretical formulations, such as sample-level personalized differential privacy, to derive tight (ϵ_c, δ_c) -DP bounds that accommodate varying user constraints without compromising the overall privacy guarantee of the aggregated model [5].

To optimize the privacy-utility trade-off under these heterogeneous conditions, novel client selec-

$$\sigma = \frac{\Delta}{\epsilon} \sqrt{2 \ln \left(\frac{1.25}{\delta} \right)}$$

Figure 21: Chart from 'DPFedBank: Crafting a Privacy-Preserving Federated Learning Framework for Financial Institutions with Policy Pillars' (arXiv:2410.13753)

tion strategies have replaced fixed uniform sampling with adaptive mechanisms like Poisson sampling and two-stage hybrid schemes. These methods leverage privacy amplification by subsampling, effectively reducing the cumulative privacy loss while minimizing communication and computation overheads. By dynamically adjusting selection probabilities according to individual budget availability and contribution importance, these frameworks mitigate the impact of stragglers and random dropouts, ensuring that the privacy accounting remains rigorous even when the available client population fluctuates unpredictably across training epochs.

Empirical validations demonstrate that hierarchical and personalized budget allocation strategies significantly outperform rigid global DP settings in terms of model accuracy and convergence speed. By slicing the total privacy budget across training rounds and weighting it by client importance, these approaches achieve utility levels comparable to central differential privacy while strictly enforcing local constraints [24]. Furthermore, fine-grained calibration of noise injection and clipping thresholds allows for efficient budget expenditure, proving that tailored privacy mechanisms can simultaneously enhance end-to-end protection guarantees and maintain high model performance in practical, large-scale deployments.

2.4.2 Record-Level and User-Level Privacy Distinctions

Record-level and user-level privacy represent distinct granularities of protection within differentially private machine learning, fundamentally altering the definition of neighboring datasets [8]. Record-level differential privacy assumes that the addition or removal of a single data point constitutes the privacy threat, making it suitable for scenarios where data contributions are independent

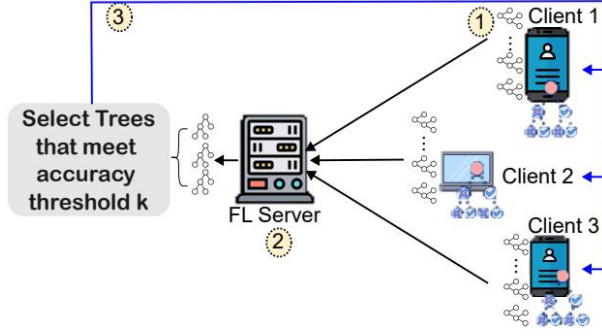


Figure 22: Chart from 'Towards Explainable Federated Learning: Understanding the Impact of Differential Privacy' (arXiv:2602.10100)

and identically distributed [25]. Conversely, user-level privacy addresses settings where a single individual contributes multiple records, such as in sequential interactions or federated learning environments. In this stricter formulation, the mechanism must obscure the presence or absence of an entire user's dataset, necessitating significantly larger noise magnitudes to bound the cumulative sensitivity across all records associated with that specific entity.

$$cf(S) = sup(S, \mathcal{D}) + \sum_{P \in MS} sup(P, \mathcal{D}),$$

Figure 23: Chart from 'Privacy-Preserving Federated Discovery of DNA Motifs with Differential Privacy' (arXiv:2304.01689)

The divergence in these definitions leads to disparate algorithmic requirements and privacy budget allocations. While record-level approaches often apply noise directly to individual gradients or statistics, user-level mechanisms require advanced composition theorems to account for the correlated nature of multi-record contributions from a single source. Consequently, achieving equivalent privacy guarantees at the user level typically demands a higher privacy budget or results in greater utility degradation compared to record-level counterparts. This distinction is critical in federated settings, where protecting the aggregate update of a client requires treating the client's local dataset as a single unit of sensitivity, rather than isolating individual training samples within that local batch.

Furthermore, the choice between these levels influences the trade-off between model utility and robustness against inference attacks. User-level privacy offers stronger protection against member-

ship inference attacks targeting specific individuals, as it prevents adversaries from determining whether a particular user participated in the training process at all [26]. However, this enhanced security often comes at the cost of slower convergence and reduced final accuracy, particularly when data distributions are non-i.i.d. across users. Recent advancements explore hierarchical budget allocation strategies to mitigate these costs, dynamically adjusting noise scales based on client importance and training phases to balance the rigorous demands of user-level confidentiality with the practical necessities of model performance.

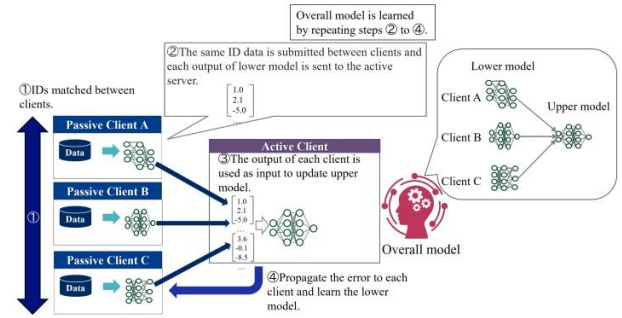


Figure 24: Chart from 'Survey of Privacy Threats and Countermeasures in Federated Learning' (arXiv:2402.00342)

2.4.3 Handling Non-IID Data with Adaptive Privacy Constraints

The integration of Differential Privacy (DP) into Federated Learning (FL) faces significant hurdles when deployed over non-IID data distributions, where statistical heterogeneity exacerbates the utility degradation caused by noise injection [27]. Standard DP mechanisms typically apply uniform noise scales across all clients, failing to account for the varying sensitivity of local gradients arising from disparate data qualities and quantities. This rigid approach often leads to severe model convergence issues, as the fixed privacy budget cannot simultaneously satisfy the strict protection requirements of sensitive minority classes and the utility needs of dominant data clusters, resulting in biased global models that perform poorly on scattered datasets.

To mitigate these challenges, recent advancements propose adaptive privacy constraints that dynamically allocate privacy budgets based on local data characteristics and gradient contributions [28]. Instead of a static noise multiplier, these frameworks employ layer-wise or client-specific

$$\mathcal{S}_{\mathcal{H}_c} \leftarrow w_{\mathcal{H}_c}(\hat{x}_c^{(i)});$$

Figure 25: Chart from 'Guarding the Middle: Protecting Intermediate Representations in Federated Split Learning' (arXiv:2602.17614)

perturbation strategies, where noise variance is calibrated according to the magnitude of weights or the KL divergence of local updates. By selectively injecting noise into less critical parameters while preserving salient features, such adaptive mechanisms optimize the privacy-utility trade-off. Furthermore, the adoption of tighter privacy accounting methods, such as Rényi Differential Privacy, allows for more precise composition of multiple training rounds, enabling stricter cumulative privacy guarantees without disproportionately sacrificing model accuracy in heterogeneous environments [24].

$$\mathcal{L}_{\text{PDP}} = \mathcal{L} + \kappa \sum_i \theta_i^2 x_i^2$$

Figure 26: Chart from 'Differential Privacy: Gradient Leakage Attacks in Federated Learning Environments' (arXiv:2510.23931)

Emerging solutions also explore hybrid architectures that combine adaptive DP with secure multi-party computation to address the dual threats of inference attacks and data imbalance. These approaches facilitate personalized privacy levels, allowing individual clients to define their own privacy parameters while maintaining a unified perturbation algorithm for global aggregation. By decoupling the privacy budget from a one-size-fits-all constraint, the system can better accommodate the diverse regulatory and sensitivity requirements inherent in cross-silo scenarios, such as those found in healthcare and finance. Consequently, adaptive constraint handling not only enhances robustness against gradient leakage but also ensures stable convergence and fair performance across diverse, non-IID data partitions.

3 Communication-Efficient and Wireless-Integrated Privacy-Preserving Federated Learning

3.1 Over-the-Air Computation and Channel Utilization

3.1.1 Exploiting Wireless Noise for Natural Privacy

Exploiting the inherent stochasticity of wireless channels offers a unique paradigm for achieving natural privacy in federated learning systems without relying solely on synthetic noise injection [3]. Unlike traditional approaches that mandate the generation of artificial Gaussian or Laplace noise to satisfy differential privacy constraints, this method leverages the additive nature of thermal noise and fading effects present during uplink transmission. By carefully calibrating transmission power and utilizing the superposition property of multiple access channels, the aggregate signal received at the base station inherently contains perturbed gradients. This physical-layer privacy mechanism effectively masks individual client updates, providing a baseline level of protection that complements algorithmic safeguards while reducing the computational overhead associated with local noise generation.

The integration of wireless noise into privacy frameworks necessitates a rigorous analysis of the privacy-utility trade-off, particularly when combined with techniques like gradient sparsification and adaptive clipping [10]. Research indicates that the randomness introduced by the channel can amplify privacy guarantees, potentially allowing for a reduction in the intensity of artificially added noise while maintaining equivalent privacy bounds. However, the variable nature of channel conditions introduces challenges in maintaining consistent privacy levels across communication rounds. Consequently, advanced scheduling algorithms and resource block allocation strategies are employed to optimize the signal-to-noise ratio, ensuring that the natural noise contribution remains sufficient to obscure sensitive information without degrading model convergence or introducing excessive utility loss.

Furthermore, recent advancements extend this concept by analyzing privacy leakage propagation through network topologies, where the cumulative effect of noisy mechanisms along transmis-

sion paths enhances information degradation. Theoretical bounds demonstrate that privacy amplification occurs not only through local noise addition but also via the delay and distortion inherent in multi-hop wireless communications. While current implementations primarily focus on Gaussian mechanisms due to their favorable composition properties, future directions explore hybrid approaches that combine channel-induced noise with adaptive differential privacy budgets. This synergy aims to dynamically adjust noise scales based on real-time channel states and training performance, offering a robust, energy-efficient solution for privacy-preserving learning in resource-constrained wireless environments.

3.1.2 Power Allocation and Beamforming for Privacy Optimization

Power allocation and beamforming serve as critical physical-layer mechanisms for enhancing privacy in federated learning by manipulating signal-to-interference-plus-noise ratios at potential eavesdroppers. By dynamically adjusting uplink transmit powers, systems can ensure that the received signal quality at unauthorized nodes remains below a decoding threshold while maintaining sufficient fidelity for legitimate aggregation. This approach leverages the stochastic nature of wireless channels to introduce inherent uncertainty, effectively acting as a natural privacy amplifier that complements artificial noise injection without incurring additional communication overhead.

Beamforming strategies further refine this protection by spatially directing signal energy toward the intended parameter server while nulling interference in the direction of adversaries. Advanced optimization frameworks jointly design beamforming vectors and power control coefficients to minimize the total privacy leakage, often formulated as minimizing the mutual information between transmitted gradients and intercepted signals. These techniques are particularly effective in multi-antenna scenarios, where degrees of freedom allow for precise spatial filtering that degrades an attacker's ability to reconstruct sensitive local updates even when channel state information is partially known.

The integration of these physical-layer techniques with differential privacy creates a synergistic effect, allowing for a reduction in the magnitude of artificially injected Gaussian noise required to meet specific privacy budgets [4]. Op-

timization problems in this domain typically balance the trade-off between convergence speed, represented by the optimality gap, and the cumulative privacy loss over communication rounds [29]. By solving for locally optimal scheduling and resource allocation, systems can achieve rigorous privacy guarantees with minimal degradation in model utility, demonstrating that strategic resource management is as vital as cryptographic perturbations in securing distributed learning environments.

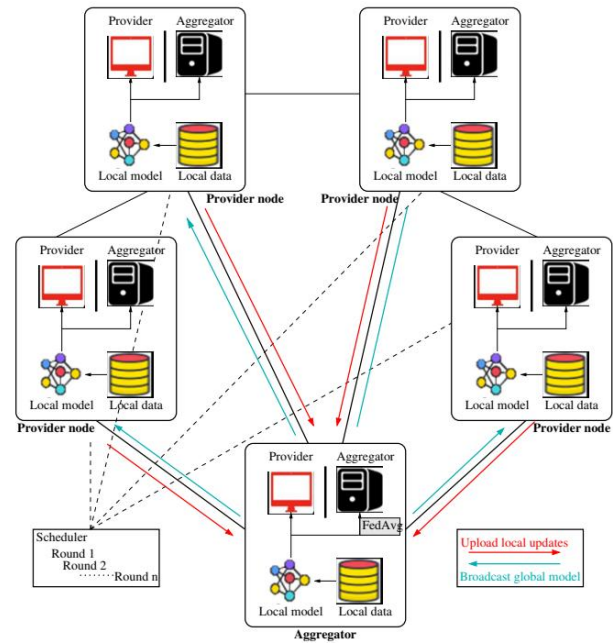


Figure 27: Chart from 'Using Decentralized Aggregation for Federated Learning with Differential Privacy' (arXiv:2311.16008)

3.1.3 Analog Aggregation under Fading Channel Constraints

Analog aggregation in wireless federated learning leverages the superposition property of fading multiple-access channels to compute weighted sums of model updates directly over the air. However, stochastic channel fading introduces significant distortion, as the effective aggregation weights become coupled with time-varying channel gains. To maintain aggregation fidelity, transmitters must employ channel inversion power control, scaling their signals by the inverse of the channel magnitude. This approach, while theoretically sound for aligning phases and magnitudes, imposes severe peak power constraints on devices experiencing deep fades, often rendering strict inversion infeasible in practical energy-limited sce-

narios.

The integration of differential privacy further complicates this physical layer operation [5]. Unlike digital aggregation where noise is added precisely post-computation, analog schemes require clients to inject Gaussian noise locally prior to transmission to preserve privacy against the server or eavesdroppers. The cumulative noise variance at the receiver is a function of both the intentionally added privacy noise and the amplified thermal noise resulting from channel inversion. Consequently, the privacy-utility trade-off becomes intrinsically linked to channel statistics; users with poor channel conditions must transmit at higher powers to overcome fading, inadvertently amplifying background noise and degrading the global model's convergence rate beyond the theoretical bounds established for ideal channels.

Recent advancements address these constraints through joint optimization of user scheduling, power allocation, and privacy budget distribution. Rather than enforcing uniform participation, adaptive schemes selectively silence users whose channel gains fall below a specific threshold to prevent excessive noise amplification. Furthermore, dynamic adjustment of local noise variances allows the system to balance the required differential privacy guarantees with the available signal-to-noise ratio [10]. By treating the fading channel not merely as a impairment but as a resource constraint within the privacy mechanism design, these methods achieve robust convergence while satisfying rigorous privacy standards under realistic wireless conditions.

3.2 Resource-Constrained Optimization and Quantization

3.2.1 Bit-Efficient Gradient Compression Techniques

Bit-efficient gradient compression techniques primarily leverage sparsification and quantization to minimize communication overhead while preserving differential privacy guarantees. Sparsifiers operate by transmitting only a subset of gradient elements, effectively setting insignificant values to zero. This approach inherently reduces the sensitivity of the transmitted messages, thereby facilitating privacy amplification through subsampling. Unlike methods that compress Hessian matrices, modern frameworks focus on compressing gradient vectors to further lower bandwidth require-

ments without sacrificing the convergence properties essential for non-convex optimization in federated settings.

Quantization methods map high-precision gradients to low-bit representations, though they introduce unique challenges regarding sensitivity and noise distribution. While standard quantizers can inadvertently increase gradient sensitivity and disrupt the Gaussian perturbation required for rigorous privacy accounting, stochastic quantization schemes offer a viable alternative. These approaches allow for a tunable trade-off where lower bit levels can theoretically enhance privacy protection by increasing inherent randomness, provided that the clipping thresholds are carefully managed to prevent bias. However, improper quantization can distort the noise distribution, complicating the theoretical analysis of privacy bounds compared to full-precision counterparts [30].

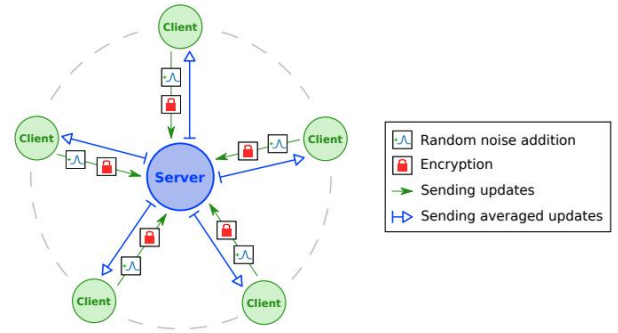


Figure 28: Chart from 'Protecting Data from all Parties: Combining Homomorphic Encryption and Differential Privacy in Federated Learning' (arXiv:2205.04330)

To address the limitations of static compression, adaptive strategies dynamically adjust clipping thresholds and bit allocations based on the evolving gradient distribution. By aligning compression parameters with benign gradient characteristics, these methods mitigate the accuracy degradation often caused by aggressive noise addition or excessive information loss. Furthermore, integrating contractive compression with partial participation enables second-order optimization methods to achieve superior communication efficiency. Ultimately, the synergy between bit-efficient encoding and adaptive clipping ensures that privacy budgets are utilized optimally, maintaining model utility even under strict communication and privacy constraints.

3.2.2 Joint Design of Communication and Privacy Parameters

The joint design of communication and privacy parameters addresses the intrinsic trade-off between transmission efficiency and data protection in federated learning [23]. Traditional approaches often treat noise injection for differential privacy and sparsification for communication compression as independent processes, leading to suboptimal utility. However, recent theoretical advancements demonstrate that inherent sparsification characteristics can bolster privacy protection through subsampling amplification. By modeling the interaction between the number of transmitted coordinates and the variance of injected noise, it becomes possible to derive tighter privacy bounds where increased communication efficiency paradoxically reduces convergence error, contrary to standard assumptions in non-private settings.

To operationalize this synergy, optimization frameworks are formulated to simultaneously minimize total privacy leakage and convergence bounds under fixed system constraints. These formulations treat privacy mechanisms, such as noise variance and clipping bounds, alongside communication variables like compression ratios and transmission probabilities as coupled decision variables. The server solves this joint optimization problem to determine optimal parameter configurations that satisfy specific (ϵ, δ) -differential privacy criteria while maintaining target accuracy levels. This approach often leverages advanced accounting tools, including Gaussian Differential Privacy and Rényi Differential Privacy, to precisely quantify the cumulative privacy loss over multiple communication rounds without overly conservative estimates [31].

Furthermore, dynamic allocation strategies enable adaptive adjustment of privacy budgets and noise scales across training iterations based on real-time convergence metrics [24]. Instead of employing a static privacy budget, these methods guide clients to apply varying levels of Gaussian noise or adjust quantization levels dynamically, ensuring robust protection during sensitive early training phases while preserving model fidelity in later stages. Experimental evidence suggests that such adaptive joint design can significantly reduce the effective privacy budget consumption-sometimes by substantial margins-while maintaining comparable model accuracy to

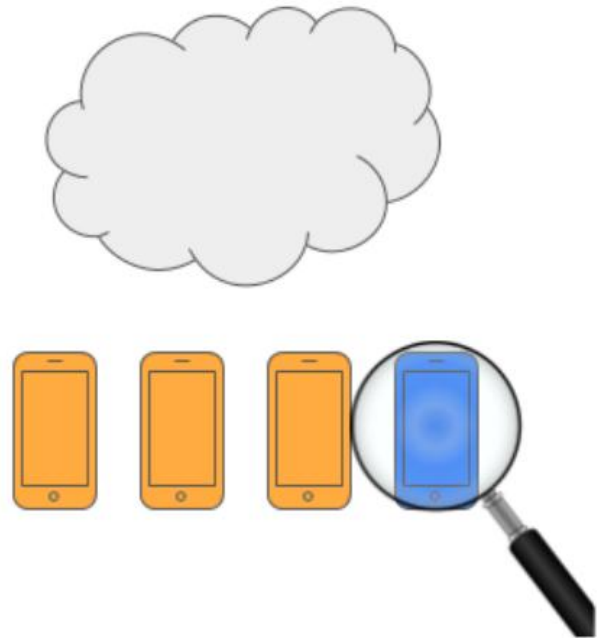


Figure 29: Chart from 'Understanding Unintended Memorization in Federated Learning' (arXiv:2006.07490)

fixed-budget baselines [7]. This holistic parameter tuning effectively transforms the privacy-utility-communication triad from a zero-sum game into a coordinated optimization landscape.

3.2.3 Energy-Aware Scheduling for Mobile Edge Devices

Energy-aware scheduling in mobile edge computing prioritizes the mitigation of resource constraints inherent to battery-powered devices while maintaining collaborative learning efficiency. Given the heterogeneity of edge hardware, scheduling mechanisms must dynamically balance computational load and communication frequency to prevent premature device exhaustion. Strategies often involve adaptive selection of participating clients based on real-time energy levels and channel conditions, ensuring that only devices with sufficient power reserves engage in local training rounds. This approach minimizes the risk of stragglers caused by energy depletion and extends the operational lifespan of the distributed network without significantly compromising model convergence rates.

To achieve optimal trade-offs between energy consumption and learning performance, advanced algorithms integrate dynamic voltage and frequency scaling (DVFS) with communication compression techniques. By adjusting the processor

frequency and transmission power according to the complexity of local updates and current battery status, these schedulers reduce the energy cost per iteration. Furthermore, incorporating quantization and sparsification into the transmission protocol lowers the data volume exchanged between edge nodes and the central server, directly reducing radio energy expenditure. Such joint optimization frameworks often formulate the scheduling problem as a convex optimization task, solving for the ideal number of local epochs and transmission intervals that satisfy strict energy budgets while adhering to differential privacy requirements.

Recent developments also explore decentralized scheduling policies that eliminate the overhead of centralized coordination, thereby further conserving network energy. These methods leverage local observations of energy harvest rates and task urgency to make autonomous participation decisions, fostering scalability in large-scale Internet of Things deployments. By embedding energy constraints directly into the objective function of federated optimization, modern schedulers ensure fairness across devices with varying power capabilities. Consequently, energy-aware scheduling transforms mobile edge devices from passive data sources into intelligent, self-regulating participants capable of sustaining long-term collaborative learning tasks under stringent power limitations.

3.3 Hierarchical and Decentralized Network Architectures

3.3.1 Multi-Tier Aggregation with Intermediate Servers

Multi-tier aggregation architectures introduce intermediate servers to mitigate communication bottlenecks and enhance scalability in large-scale federated learning deployments [32]. In this hierarchical framework, edge devices transmit local model updates to nearby intermediate aggregators rather than a distant central server. These intermediate nodes perform preliminary aggregation, often utilizing protocols like federated averaging or robust statistical methods, before forwarding compressed global updates to the top-tier server. This structure significantly reduces latency and bandwidth consumption, particularly in wireless edge networks where direct connectivity to a central master is constrained by distance or limited spectrum availability.

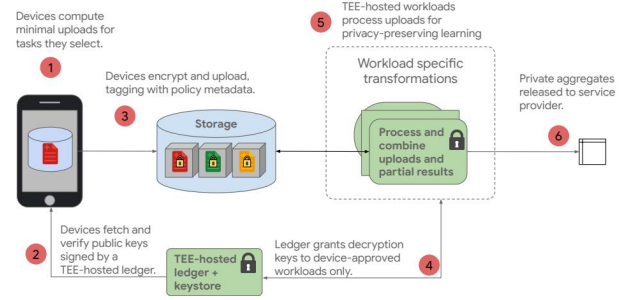


Figure 30: Chart from 'Federated Learning in Practice: Reflections and Projections' (arXiv:2410.08892)

The integration of intermediate layers also facilitates advanced privacy preservation and robustness mechanisms against adversarial attacks. By distributing the aggregation workload, the system can implement localized differential privacy noise addition or secure multi-party computation at the edge tier, preventing the central server from accessing individual client contributions. Furthermore, intermediate servers can act as filters for Byzantine faults, employing robust aggregation rules such as median or trimmed mean to discard malicious updates before they propagate to the global model [33]. This layered defense strategy ensures that the final global model maintains high utility while satisfying strict privacy budgets and resisting poisoning attempts from compromised nodes.

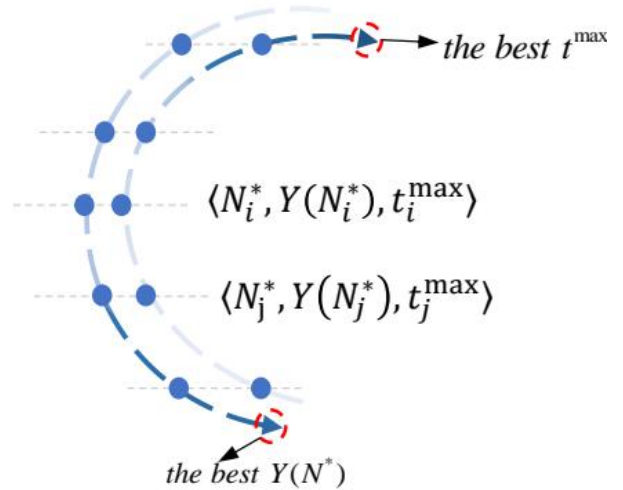


Figure 31: Chart from 'CoSIFL: Collaborative Secure and Incentivized Federated Learning with Differential Privacy' (arXiv:2509.23190)

However, designing efficient multi-tier systems requires careful optimization of resource allocation, client scheduling, and synchronization pro-

protocols across heterogeneous network conditions. Dynamic graph structures necessitate adaptive transition matrices to handle time-varying connectivity and partial client participation without compromising convergence guarantees. Recent advancements explore overlapping group strategies and cooperative mobile edge networking to balance load distribution and minimize straggler effects. Despite the added complexity in coordination, multi-tier aggregation with intermediate servers represents a critical evolution for deploying federated learning in real-world, geographically dispersed environments where single-tier architectures fail to meet performance and privacy requirements [32].

3.3.2 Decentralized Consensus in Wireless Mesh Networks

Decentralized consensus in wireless mesh networks eliminates the single point of failure inherent in centralized federated learning by enabling peer-to-peer model aggregation through gossip-based or push-sum protocols. In this architecture, nodes iteratively update local parameters by averaging models with immediate neighbors, governed by time-varying transition matrices that accommodate dynamic network topologies and partial node participation. This pairwise observability constraint fundamentally alters the privacy landscape, as attackers can only access local models involved in specific communication rounds rather than the global state, thereby enabling privacy amplification by decentralization.

To safeguard against honest-but-curious neighbors or external eavesdroppers, these consensus mechanisms frequently integrate Local Differential Privacy (LDP) directly into the gradient exchange process. By injecting calibrated Gaussian or Laplacian noise into local updates prior to transmission, the system ensures that individual data contributions remain indistinguishable while maintaining convergence guarantees for strongly convex and non-convex objectives. Advanced variants employ adaptive noise allocation and subsampling techniques to amplify privacy guarantees, leveraging the stochastic nature of wireless channel access and asynchronous update schedules to reduce the cumulative privacy loss over multiple communication rounds without significantly degrading model utility.

Furthermore, recent developments address the dual challenges of communication efficiency and

Byzantine robustness within these decentralized frameworks. Techniques such as gradient clipping, centered momentum, and verifiable secret sharing are employed to mitigate the impact of malicious nodes attempting to poison the global model, ensuring resilient convergence even under adversarial conditions. The integration of over-the-air computation and interference management strategies further optimizes bandwidth utilization in dense mesh environments, allowing for scalable, low-latency consensus that aligns with the stringent resource constraints of wireless edge devices while preserving rigorous differential privacy bounds.

3.3.3 Topology-Aware Privacy Propagation Mechanisms

Topology-aware privacy propagation mechanisms leverage the structural properties of decentralized networks to optimize the privacy-utility trade-off in federated learning [3]. Unlike traditional approaches that apply uniform noise regardless of network configuration, these mechanisms analyze how privacy leakage propagates across specific graph topologies, such as rings or 2D tori. By characterizing the relationship between node connectivity and information flow, algorithms can dynamically adjust noise injection strategies based on the distance between workers and the number of intermediate noisy mechanisms encountered along communication paths.

The theoretical foundation of these mechanisms relies on bounding privacy loss through iterative structures induced by random walks and local updates. Advanced frameworks employ f -differential privacy and zero-concentrated differential privacy to derive tighter bounds that account for mixture mechanisms and propagation delays [5]. This analysis reveals an inverse relationship between privacy leakage and graph distance, where increased separation between nodes results in degraded information quality due to cumulative noise. Consequently, privacy amplification occurs naturally as data traverses longer paths, allowing for reduced initial noise intensity while maintaining rigorous global privacy guarantees [16].

Furthermore, these mechanisms address the challenges of overlapping group structures by quantifying leakage between arbitrary pairs of workers rather than relying on a single global bound. This granular approach enables the optimization of artificial noise and resource scheduling to minimize total privacy leakage without com-

promising model convergence. By integrating topology-dependent error bounds, recent advancements demonstrate that higher communication efficiency can simultaneously reduce convergence errors and enhance privacy protection, offering a robust solution for heterogeneous and dynamic decentralized environments.

3.4 Dynamic Privacy Budget Management

3.4.1 Time-Adaptive Noise Scaling Strategies

Time-adaptive noise scaling strategies address the inherent tension between privacy guarantees and model utility by dynamically adjusting noise variance across training iterations. Unlike static approaches that apply a constant noise multiplier, these methods leverage the observation that early-stage updates tolerate higher perturbation levels without significantly impacting final convergence, whereas later stages require finer granularity to stabilize around optima. By allocating larger variances during initial rounds and geometrically decreasing them over time, algorithms can satisfy cumulative privacy budgets more efficiently while mitigating the divergence often caused by excessive noise in critical convergence phases [29].

Advanced implementations utilize contraction effects and repeated noise injection to derive tighter trade-off functions, particularly within the framework of f -differential privacy [4]. This allows for precise calibration where the noise standard deviation σ_t is scaled relative to the clipping bound and the remaining privacy budget at iteration t [10]. Such mechanisms exploit the fact that noise added in earlier steps has a diminished effect on the total error at the final iteration due to telescoping properties in the optimization trajectory. Consequently, the scheduler can aggressively reduce noise magnitude as training progresses, ensuring that the aggregate privacy loss remains within (ϵ, δ) bounds while maximizing the signal-to-noise ratio for sensitive parameter updates near convergence.

Furthermore, these strategies often integrate adaptive clipping thresholds that evolve alongside the noise scale to handle non-convex loss landscapes and heterogeneous client data. By jointly optimizing the clipping bound and noise variance based on gradient norm distributions or loss fluctuations, the system minimizes the residual error terms in convergence bounds. This dual adaptation ensures robustness against user drift and pre-

vents the optimization process from stalling, effectively balancing the competing demands of rigorous privacy protection and high test accuracy in federated and distributed learning environments.

3.4.2 Convergence-Privacy Trade-off Analysis

The convergence-privacy trade-off in differentially private federated learning is fundamentally governed by the magnitude of injected noise relative to the optimization landscape [3]. Theoretical analyses establish that while differential privacy guarantees are strengthened by increasing noise variance or tightening the privacy budget ϵ , this directly inflates the upper bound on the optimality gap [34]. Consequently, excessive noise accumulation over communication rounds can destabilize the gradient descent trajectory, leading to slower convergence rates or even divergence if the signal-to-noise ratio falls below a critical threshold required for stationary point attainment.

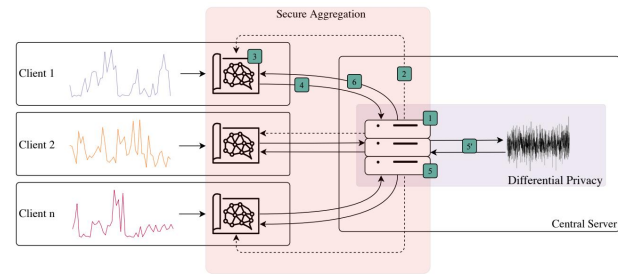


Figure 32: Chart from 'Secure Federated Learning for Residential Short Term Load Forecasting' (arXiv:2111.09248)

Recent advancements address this tension by moving beyond fixed privacy budgets toward dynamic allocation strategies that adapt noise variance across training iterations [24]. By leveraging composition theorems and privacy amplification via subsampling, these methods optimize the distribution of the total privacy budget to minimize utility loss while maintaining rigorous (ϵ, δ) -differential privacy guarantees [28]. Crucially, incorporating gradient clipping into the convergence analysis allows for bounded sensitivity, ensuring that the optimization error induced by privacy mechanisms remains independent of the total number of iterations under smooth objective functions, thereby preventing unbounded error growth.

Furthermore, the trade-off is intricately linked to hyperparameters such as the clipping threshold and client sampling rates, which jointly influ-

ence both privacy leakage and convergence speed. Tighter clipping reduces sensitivity and required noise but introduces bias that may hinder accuracy, whereas aggressive sampling enhances privacy amplification but potentially slows convergence due to reduced effective batch sizes. Optimal performance is achieved by numerically solving for parameters that minimize the combined objective of privacy loss and convergence error, often revealing that adaptive schemes significantly outperform static configurations in balancing robust privacy protection with efficient model utility.

3.4.3 Fairness-Aware Resource Distribution Protocols

Fairness-aware resource distribution protocols in federated learning increasingly integrate differential privacy (DP) mechanisms to balance equitable model performance with rigorous data protection [23]. Traditional approaches often suffer from inefficient privacy budget allocation, where uniform distribution leads to unnecessary utility loss for specific client groups. To address this, recent frameworks introduce adaptive DP strategies that dynamically adjust noise variance based on individual client constraints and contribution levels. By minimizing optimization error under predefined privacy budgets, these protocols ensure that resource-constrained devices are not disproportionately penalized, thereby maintaining convergence accuracy while preserving strict privacy guarantees across heterogeneous networks.

Advanced methodologies further refine this balance by employing fine-grained privacy notions tailored to decentralized topologies, such as Pair-wise Network f -DP. This extension quantifies privacy loss between specific user pairs during random-walk communication, enabling tight accounting for both user-level and record-level guarantees. Such precision allows for the implementation of fairness-aware algorithms like FedFair, which incorporate novel loss functions to simultaneously optimize global model utility and inter-group fairness. By leveraging structured noise injection via shared secrets or correlated perturbations, these systems mitigate the trade-off between communication efficiency and privacy, ensuring that sparsification techniques amplify rather than degrade protection levels.

Moreover, robust protocols now incorporate verifiable secure aggregation to prevent multi-round privacy leakage while defending against Byzan-

tine attacks [35]. These systems utilize secure distributed noise generation to guarantee the integrity of added perturbations, ensuring that the aggregator cannot access intermediate global models without compromising the differential privacy shield. The integration of geometry-aligned noise strategies further enhances location privacy in edge computing scenarios by aligning spatial distributions with sensitivity requirements. Consequently, modern fairness-aware distribution protocols achieve a superior privacy-utility trade-off, enabling scalable, resilient, and equitable federated learning deployments across diverse and potentially adversarial environments [36].

$$\mathbf{y}_i = \mathbf{x}_i + \sum_{i < j} \text{PRG}(s_{i,j}) - \sum_{i > j} \text{PRG}(s_{j,i})$$

Figure 33: Chart from 'Efficient Dropout-resilient Aggregation for Privacy-preserving Machine Learning' (arXiv:2203.17044)

$$\Delta := \sum_i^m \Delta_i \mod M,$$

Figure 34: Chart from 'Federated Learning of Gboard Language Models with Differential Privacy' (arXiv:2305.18465)

4 Hybrid Cryptographic and Multi-Party Computation Frameworks for Federated Learning

4.1 Homomorphic Encryption and Secure Computation

4.1.1 Partial and Fully Homomorphic Encryption Schemes

Partial Homomorphic Encryption (PHE) schemes, particularly those supporting additive operations, serve as the foundational backbone for many privacy-preserving machine learning protocols due to their computational efficiency [37]. By leveraging mathematical constructs such as the Learning With Errors (LWE) problem, these schemes enable the secure aggregation of model updates or the evaluation of linear functions directly on ciphertexts without decryption. In federated learning contexts, additive homomorphism

allows servers to compute global model parameters from encrypted local gradients, ensuring that sensitive data remains confidential throughout the transmission and aggregation phases. However, the utility of PHE is inherently restricted to specific algebraic structures, preventing the evaluation of complex non-linear activation functions or deep neural network architectures without resorting to expensive interaction protocols or approximation techniques.

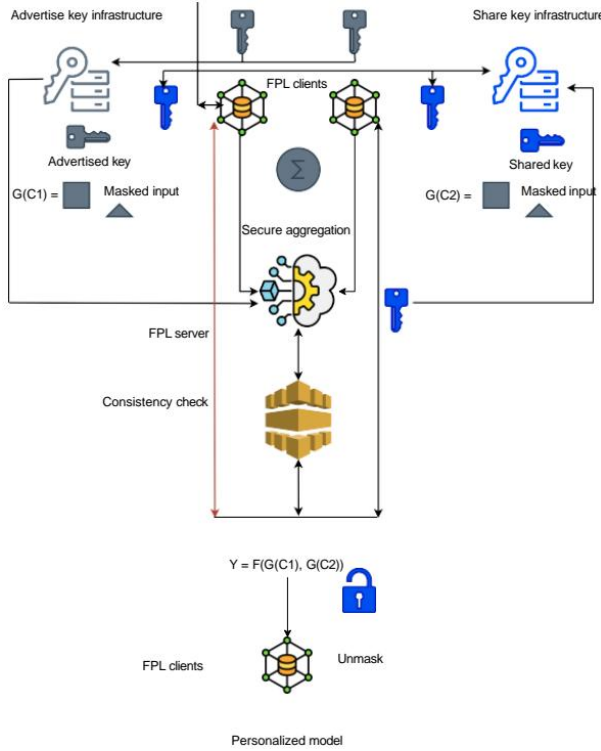


Figure 35: Chart from 'Privacy Preserving Machine Learning Model Personalization through Federated Personalized Learning' (arXiv:2505.01788)

Fully Homomorphic Encryption (FHE) overcomes these structural limitations by permitting the evaluation of arbitrary functions on encrypted data, theoretically enabling complete model training and inference in an untrusted environment [38]. Since Gentry's seminal construction, FHE schemes have evolved to manage the accumulation of noise introduced during homomorphic multiplications through bootstrapping procedures, which refresh ciphertexts to allow for unlimited computational depth. Despite this theoretical versatility, practical deployment of FHE faces significant hurdles regarding latency and memory overhead, as operations on ciphertexts are orders of magnitude slower than plaintext computations. Consequently, current implementations often rely on fixed-point

arithmetic representations and optimized polynomial encoding strategies to mitigate performance bottlenecks while maintaining the rigorous security guarantees provided by ideal lattice-based cryptography.

$$\mu = \sqrt{\sum_i \mu_i^2}$$

Figure 36: Chart from 'Considerations on the Theory of Training Models with Differential Privacy' (arXiv:2303.04676)

To balance the trade-off between the efficiency of PHE and the expressiveness of FHE, recent research has increasingly focused on hybrid architectures and Somewhat Homomorphic Encryption (SHE) schemes tailored for specific deep learning workloads [30]. These approaches strategically limit the multiplicative depth to match the requirements of particular neural network layers, avoiding the heavy cost of full bootstrapping while still supporting necessary non-linear operations via secure multi-party computation or polynomial approximations. Furthermore, integrating these cryptographic primitives with differential privacy mechanisms offers a robust defense against inference attacks, providing a layered security model where cryptographic secrecy protects data in transit and during processing, while statistical noise guards against information leakage from the final model outputs [39].

4.1.2 Secure Multi-Party Computation for Gradient Aggregation

Secure Multi-Party Computation (MPC) serves as a foundational cryptographic primitive for gradient aggregation in federated learning, enabling mutually distrusting clients to jointly compute a global model update without revealing individual contributions [37]. By leveraging protocols such as secret sharing and oblivious transfer, MPC ensures that the central aggregator receives only the final summation of gradients, effectively isolating raw local updates from both the server and poten-

tial eavesdroppers. This approach eliminates the need for a trusted third party, as security relies on the assumption that a threshold of participants remains honest, thereby preventing the reconstruction of any single client's input during the collaborative computation process.

The implementation of MPC for secure aggregation typically involves clients splitting their gradient vectors into random shares distributed among peers or designated non-colluding servers. Through secure summation protocols, these shares are aggregated locally before being transmitted, ensuring that intermediate values reveal no information about the underlying data. While this method provides rigorous privacy guarantees against gradient leakage and inversion attacks, it introduces significant communication overhead and strict synchronization requirements. Consequently, the feasibility of MPC in large-scale, heterogeneous networks is often constrained by bandwidth limitations and the latency associated with multiple rounds of interaction required to maintain cryptographic integrity.

To address these scalability challenges, recent advancements integrate MPC with differential privacy, creating hybrid frameworks that balance computational efficiency with robust privacy preservation. In such systems, clients may apply local noise mechanisms to their shares before aggregation, leveraging the post-processing properties of differential privacy to amplify anonymity without incurring additional privacy costs during momentum-based optimization. Although pure MPC solutions offer strong theoretical security, the practical deployment in resource-constrained environments increasingly favors optimized secure summation algorithms that reduce pairwise communication complexity while maintaining resistance against semi-honest adversaries and ensuring the confidentiality of sensitive model updates.

4.1.3 Functional Encryption for Selective Data Disclosure

Functional Encryption (FE) represents a paradigm shift from traditional all-or-nothing encryption by enabling fine-grained access control over encrypted data within federated learning architectures [40]. Unlike homomorphic encryption, which permits arbitrary computation on ciphertexts, FE allows a holder of a specific secret key to learn only a predetermined function of the un-

derlying plaintext while keeping all other information theoretically hidden. This capability is particularly critical for selective data disclosure, where the goal is to reveal only aggregated model updates or specific statistical features to a central server without exposing individual client gradients or raw sensitive attributes.

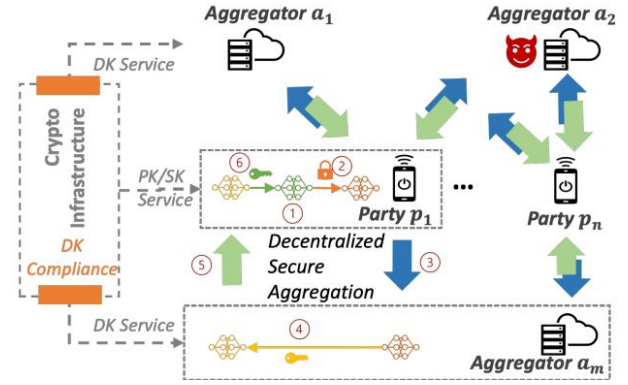


Figure 37: Chart from 'TAPFed: Threshold Secure Aggregation for Privacy-Preserving Federated Learning' (arXiv:2501.05053)

In the context of privacy-preserving machine learning, FE schemes facilitate selective disclosure by binding decryption capabilities to specific predicates or functions defined during the key generation phase. When applied to federated systems, this ensures that even a curious aggregator cannot reconstruct individual contributions beyond the intended output of the aggregation function. Recent advancements have integrated FE with differential privacy mechanisms to create hybrid frameworks; here, FE provides cryptographic guarantees against unauthorized access, while differential privacy bounds the information leakage resulting from the disclosed function outputs, thereby mitigating risks associated with inference attacks and model inversion.

Despite its theoretical robustness, the deployment of FE for selective disclosure faces significant practical hurdles, primarily concerning computational overhead and key management complexity. Constructing efficient FE schemes for complex functions often incurs substantial latency during encryption and key derivation, which can impede scalability in large-scale, dynamic federated environments. Current research directions focus on optimizing pairing-based constructions and exploring lattice-based alternatives to reduce ciphertext expansion and processing time, aiming to balance the stringent requirements of selective

transparency with the efficiency demands of real-world distributed learning systems.

4.2 Trusted Execution Environments and Hardware Security

4.2.1 Enclave-Based Model Training and Aggregation

Enclave-based model training leverages Trusted Execution Environments (TEEs) to establish hardware-rooted trust boundaries within federated learning architectures. By isolating computation and memory access from the host operating system, TEEs enable clients to perform local stochastic gradient descent on sensitive datasets without exposing raw data or intermediate gradients to potentially compromised infrastructure. This approach mitigates side-channel vulnerabilities inherent in software-only solutions, ensuring that the confidentiality of proprietary data is maintained even when the underlying physical servers are untrusted or managed by adversarial entities.

During the aggregation phase, secure enclaves serve as verified trusted aggregators that decrypt and combine model updates from distributed participants within a protected memory region. Unlike cryptographic secure multi-party computation which often incurs prohibitive communication overhead, enclave-based aggregation allows the central server to compute global model averages efficiently while remaining oblivious to individual client contributions. The integrity of this process is guaranteed through remote attestation protocols, which allow participating nodes to cryptographically verify the authenticity of the aggregation code before transmitting their encrypted model parameters, thereby preventing malicious injection or tampering [2].

Despite these advantages, deploying enclave-based systems introduces challenges related to memory constraints and potential leakage through access patterns. The limited enclave memory size often necessitates sophisticated model partitioning strategies or iterative swapping mechanisms, which can degrade training throughput compared to standard plaintext execution. Furthermore, while TEEs protect data content, they do not inherently obscure memory access traces, requiring supplementary techniques such as oblivious RAM or differential privacy to fully harden the system against sophisticated inference attacks targeting execution metadata.

4.2.2 Verifiable Computing and Integrity Proofs

Verifiable computing mechanisms address the critical challenge of ensuring computational integrity in federated learning, particularly when the central server operates under an honest-but-curious or fully malicious adversary model [30]. By leveraging cryptographic primitives such as Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge (zk-SNARKs) and Zero-Knowledge Scalable Transparent Arguments of Knowledge (zk-STARKs), clients can generate succinct proofs attesting to the correctness of their local model updates without revealing underlying private data [41]. These non-interactive proofs allow the aggregator to verify that the submitted gradients result from valid training procedures on legitimate datasets, effectively mitigating risks associated with data poisoning, backdoor insertion, and model manipulation while preserving the confidentiality of individual contributions.

$$J_a(\mathcal{M}_\theta^a, D_a) = \frac{1}{n_a} \sum_{(X_i, Y_i) \in D_a} \mathcal{L}(\mathcal{M}_\theta^a(X_i), Y_i).$$

Figure 38: Chart from 'A Privacy-Preserving and Trustable Multi-agent Learning Framework' (arXiv:2106.01242)

To further strengthen system robustness, recent architectures integrate verifiable secure aggregation protocols that combine homomorphic encryption with integrity checks. Unlike standard secure aggregation which only guarantees privacy, verifiable variants enable the server to prove that the global model update is the exact sum of authenticated local contributions, preventing the injection of arbitrary values or the exclusion of specific clients [2]. This is often achieved through commitment schemes and digital signatures linked to blockchain-based immutable ledgers, which provide an auditable trail of all aggregation steps. Such tamper-proof records ensure that any deviation from the agreed-upon protocol is computationally infeasible to hide, thereby establishing end-to-end traceability and trust in decentralized environments without relying on a trusted third party.

Despite their strong security guarantees, the deployment of verifiable computing faces significant hurdles regarding computational overhead and communication latency. Generating and veri-

ifying cryptographic proofs, especially for complex neural network operations involving non-linear activations, imposes a heavy burden on resource-constrained edge devices. Consequently, current research focuses on optimizing proof generation through hardware accelerators, such as Trusted Execution Environments (TEEs), which offer attestation capabilities with lower latency compared to pure cryptographic solutions. Future directions aim to balance this trade-off by developing lightweight verification protocols and adaptive privacy budgets that maintain rigorous integrity assurances while ensuring the practical scalability required for large-scale federated learning deployments [8].

4.2.3 Hardware-Assisted Privacy Amplification

Hardware-assisted privacy amplification leverages trusted execution environments (TEEs) and secure processors to bridge the utility gap between local and global differential privacy. By offloading noise addition and aggregation to isolated hardware enclaves, these frameworks simulate a trusted curator without requiring explicit trust in the central server. This architectural shift allows participants to submit data with weaker local privacy guarantees, as the hardware ensures that intermediate values remain confidential until the final aggregated result is released, effectively amplifying the overall privacy budget through physical isolation rather than purely algorithmic perturbation.

Recent advancements integrate these hardware primitives with shuffle models to further strengthen privacy guarantees in federated learning settings [3]. In this paradigm, hardware modules perform anonymous shuffling of encrypted updates before decryption and aggregation, breaking the link between specific users and their contributions. This combination enables dynamic privacy allocation, where the system can adaptively adjust noise levels based on real-time sensitivity analysis and composite objectives. The hardware enforcement ensures that the shuffling process is verifiable and tamper-proof, providing rigorous protection against inference attacks that typically exploit gradient leakage in software-only implementations.

Despite the theoretical benefits, deploying hardware-assisted mechanisms introduces challenges regarding scalability and heterogeneous de-

vice compatibility. The computational overhead of secure enclaves and the latency introduced by hardware-based cryptographic operations can bottleneck large-scale distributed training, particularly when dealing with high-dimensional model parameters. Furthermore, ensuring the robustness of these systems against side-channel attacks and verifying the integrity of the hardware firmware remain critical open problems. Future research must focus on optimizing the trade-off between the strict security guarantees provided by hardware isolation and the efficiency required for practical, real-time machine learning applications.

4.3 Blockchain and Decentralized Trust Mechanisms

4.3.1 Smart Contracts for Automated Privacy Enforcement

Smart contracts serve as immutable, self-executing protocols that automate privacy enforcement within decentralized federated learning ecosystems, effectively removing the need for trusted central curators. By encoding differential privacy parameters directly into the blockchain logic, these contracts mandate that participating nodes inject calibrated noise into their local model updates prior to transmission. This architectural shift ensures that privacy guarantees, specifically adhering to local differential privacy constraints, are computationally enforced rather than relying on the voluntary compliance of data owners or the integrity of a central aggregator [42].



Figure 39: Chart from 'Federated Learning With Individualized Privacy Through Client Sampling' (arXiv:2501.17634)

The integration of smart contracts facilitates verifiable adherence to privacy policies throughout the training lifecycle, addressing vulnerabilities associated with inference attacks and malicious collusion. Through cryptographic verification mechanisms embedded in the contract code, the system

can validate that noise addition meets specified epsilon-delta bounds before accepting gradient updates for global aggregation. This automated auditing capability not only strengthens resilience against adversaries attempting to reconstruct sensitive training data but also ensures that the trade-off between model utility and privacy protection remains within predefined regulatory thresholds, such as those mandated by GDPR or HIPAA.

Furthermore, smart contracts enable dynamic adaptation of privacy budgets based on real-time threat assessments and convergence metrics, optimizing the balance between security and accuracy. In scenarios involving heterogeneous data distributions or varying trust levels among participants, the contract logic can adjust noise scales individually for each agent, thereby preserving global model performance while maintaining rigorous individual privacy standards. This programmable enforcement layer transforms privacy from a static configuration into an active, responsive component of the distributed learning infrastructure, ensuring robust protection even in adversarial environments where up to half of the participants may act maliciously.

4.3.2 Immutable Audit Trails for Federated Updates

The integration of blockchain technology into federated learning architectures addresses the critical need for immutable audit trails, ensuring the integrity and traceability of distributed model updates. By recording local gradient contributions and global aggregation events on a decentralized ledger, systems establish a tamper-proof history that prevents malicious actors from altering update sequences or injecting poisoned models without detection. This cryptographic linkage of blocks creates a verifiable lineage for the global model, allowing auditors to reconstruct the entire training trajectory and identify specific iterations where anomalies or adversarial interventions occurred.

Beyond mere record-keeping, these immutable logs facilitate rigorous accountability in incentive-compatible environments where participants are rewarded based on the quality of their contributions. Smart contracts can automatically verify digital signatures attached to each update against the stored hash values, ensuring that only authentic, unmodified gradients from registered clients influence the global model. This mechanism effectively mitigates risks associated with free-riding

and sybil attacks, as every computational effort is permanently anchored to a specific identity and timestamp, providing a robust foundation for dispute resolution and performance evaluation without relying on a trusted central authority.

Furthermore, the synergy between immutable ledgers and cryptographic privacy primitives enhances the overall security posture by separating the verification of update integrity from the inspection of update content. While secure aggregation protocols obscure individual contributions to preserve differential privacy guarantees, the blockchain layer concurrently validates the structural correctness and provenance of the aggregated results. This dual approach ensures that the system maintains high fidelity in model convergence while offering transparent, non-repudiable evidence of compliance with privacy policies, thereby bridging the gap between opaque cryptographic operations and the regulatory requirements for auditable machine learning systems.

4.3.3 Token-Based Incentives for Honest Participation

Token-based incentive mechanisms address the critical challenge of ensuring honest participation in federated learning by aligning individual rationality with collective model accuracy. Unlike traditional reputation systems that rely on historical behavior, cryptographic token schemes utilize unique, non-transferable identifiers to validate the genuineness of client contributions without compromising data privacy. These tokens, often generated via secure multi-party computation or additive secret sharing, allow the aggregator to verify that updates originate from legitimate participants while keeping the specific identity of the token owner hidden, thereby preventing Sybil attacks and free-riding behaviors in dishonest-majority settings.

The implementation of such schemes frequently leverages distributed ledger technology to create transparent, immutable records of contribution quality and reward distribution. By integrating fungible tokens compliant with standards like ERC-20, frameworks can establish decentralized marketplaces where data providers are compensated based on the verified utility of their local model updates. This economic layer not only encourages sustained engagement across heterogeneous devices but also facilitates a trustless environment where the verification of computational

integrity is decoupled from the exposure of raw data or intermediate gradients, effectively mitigating risks associated with insider threats and malicious collusion.

However, the deployment of token-based incentives introduces complex trade-offs between computational overhead, communication latency, and privacy guarantees. While the initial generation and management of random token lists incur significant costs, these are amortized over multiple training rounds since token ownership remains concealed throughout the process. Future research directions emphasize the development of lightweight verifiable update mechanisms and adaptive privacy budgets that can dynamically adjust rewards based on data sensitivity and contribution quality. Such advancements are essential to maintain scalability in large-scale deployments while ensuring robustness against sophisticated adversarial strategies like data poisoning and backdoor insertion.

4.4 Integrated Hybrid Defense Strategies

4.4.1 Combining Differential Privacy with Cryptography

The integration of Differential Privacy (DP) with cryptographic primitives addresses the inherent limitations of employing either technique in isolation within federated learning ecosystems [5]. While DP provides rigorous, quantifiable bounds on privacy loss through calibrated noise injection, it often necessitates a trusted aggregator in central models or suffers from significant utility degradation in local settings [43]. Conversely, cryptographic methods like Secure Multi-Party Computation (MPC) and Homomorphic Encryption ensure data confidentiality during transmission and aggregation but remain vulnerable to inference attacks on the final decrypted model parameters [37]. Hybrid frameworks strategically combine Local Differential Privacy (LDP) at the client side with secure aggregation protocols to mitigate these distinct threat vectors simultaneously.

$$\min_{w \in R^d} f(w) \quad \text{where} \quad f(w) = \frac{1}{n} \sum_{i=1}^n f_i(w)$$

Figure 40: Chart from 'An Empirical Study of Efficiency and Privacy of Federated Learning Algorithms' (arXiv:2312.15375)

A prominent architectural pattern in this do-

main involves coupling LDP with additive secret sharing or threshold cryptography. In such schemes, clients perturb their model updates locally to satisfy (ϵ, δ) -LDP guarantees before distributing shares among servers or peers. The cryptographic layer ensures that individual noisy updates remain hidden during the aggregation process, preventing the server from accessing raw gradients even if it deviates from the protocol. This synergy leverages the "amplification by shuffling" effect, where the anonymity provided by secure aggregation enhances the privacy guarantees of the local noise, allowing for reduced noise magnitude while maintaining the same overall privacy budget compared to standalone LDP approaches.

Formal analysis of these hybrid systems demonstrates that they achieve end-to-end privacy guarantees that are robust against both honest-but-curios aggregators and malicious colluding participants [44]. By distributing the noise generation and utilizing full-threshold secret sharing, these frameworks eliminate the single point of failure associated with trusted curators. The resulting privacy loss composition is often tighter than the sum of individual mechanisms, enabling a more favorable trade-off between model utility and privacy preservation. Consequently, combining DP with cryptography has emerged as a foundational strategy for deploying scalable, robust, and formally verified privacy-preserving machine learning in distributed environments [45].

$$\epsilon_{\text{empirical}} = \max \left(\log \frac{1 - \text{FP}}{\text{FN}}, \log \frac{1 - \text{FN}}{\text{FP}} \right)$$

Figure 41: Chart from 'Measuring Lower Bounds of Local Differential Privacy via Adversary Instantiations in Federated Learning' (arXiv:2206.09122)

4.4.2 Fairness and Privacy in Multi-Party Settings

In multi-party learning environments, the intersection of fairness and privacy introduces complex trade-offs that challenge traditional mitigation strategies. While Differential Privacy (DP) provides rigorous guarantees against inference attacks by injecting calibrated noise, this perturbation often exacerbates existing data disparities, leading to unequal model performance across demographic groups [4]. The indiscriminate application of noise can disproportionately degrade

utility for minority populations, thereby violating fairness metrics such as Demographic Parity and Equalized Odds. Consequently, achieving simultaneous compliance with strict privacy budgets and equitable outcome distributions remains a significant theoretical and practical hurdle in distributed settings.

To address these conflicting objectives, recent frameworks integrate Local DP with Secure Multi-Party Computation (MPC) to decouple privacy preservation from aggregation trust assumptions [46]. By combining gradient clipping with cryptographic secure aggregation, these approaches aim to minimize the noise magnitude required for a given privacy level, thus preserving model accuracy. However, the interplay between cryptographic protocols and fairness constraints is delicate; increasing privacy protections via larger noise variance or stricter epsilon values frequently disrupts the convergence of fair models. Empirical studies suggest that while privacy mechanisms effectively obscure individual contributions, they may inadvertently encode biases into the global model if the underlying local data distributions are highly heterogeneous [47].

$$\text{AvgRank}(m) = \frac{1}{|\mathcal{P}||\mathcal{E}|} \sum_{p \in \mathcal{P}} \sum_{\varepsilon \in \mathcal{E}} \text{Rank}_m(p, \varepsilon),$$

Figure 42: Chart from 'Federated Survival Analysis with Node-Level Differential Privacy: Private Kaplan-Meier Curves' (arXiv:2509.00615)

$$\epsilon \geq c 2^{\frac{q \sqrt{T \log(1/\delta)}}{\sigma}}$$

Figure 43: Chart from 'Gradient-Leakage Resilient Federated Learning' (arXiv:2107.01154)

Advanced solutions propose phase-wise training methodologies and adaptive noise injection strategies to navigate this tri-lemma of privacy, fairness, and utility. These methods dynamically adjust privacy parameters based on local fairness metrics, allowing for customized protection levels that mitigate disparate impact without compromising overall security guarantees. Furthermore, formal verification techniques are increasingly employed to quantify the privacy loss specif-

ically regarding sensitive attributes, ensuring that the post-processing properties of DP do not erode fairness constraints. Despite these advancements, optimizing the balance requires careful calibration of hyperparameters, as enhancing one dimension often necessitates a measurable sacrifice in the others, demanding comprehensive framework designs that account for scalability and robustness alongside ethical considerations.

4.4.3 Resilience Against Collusion and Insider Threats

Resilience against collusion in decentralized learning primarily relies on the information-theoretic security provided by Additive Secret Sharing (ASS). In this framework, individual client updates are split into shares distributed among m intermediate servers. As long as an adversary compromises fewer than all m servers, strictly observing a subset of shares reveals zero information about the underlying private data. This threshold property ensures that even if up to $f < m$ intermediate servers are corrupted or if curious clients attempt to isolate specific contributions, the privacy of individual updates remains intact, provided the aggregation process does not expose the full set of shares simultaneously.

However, inherent limitations exist when the threat model expands to include active adversaries or total server compromise. If an attacker controls all intermediate nodes or intercepts every share-carrying link, the aggregated secret can be reconstructed, representing a fundamental failure condition of full-threshold ASS [48]. To address scenarios involving malicious actors who may arbitrarily deviate from protocol specifications, advanced four-party protocols have been developed to maintain security with a corruption threshold of 25% [49]. Furthermore, while standard secure aggregation protects against passive eavesdropping, robust aggregation rules are increasingly integrated to identify and mitigate the impact of malicious participants attempting data poisoning or model inversion attacks [50].

Insider threats also emerge from colluding clients who may share their local noise contributions to degrade Differential Privacy guarantees. When participants coordinate to subtract their known noise from the aggregate, the effective privacy budget diminishes, potentially exposing sensitive gradients [30]. Mitigation strategies involve either increasing noise variance, which

$$\Delta f = \max_{\substack{x, y \in \mathbb{N}^{|\mathcal{X}|} \\ \|x - y\|_1 = 1}} \|f(x) - f(y)\|_1.$$

Figure 44: Chart from 'DDP-SA: Scalable Privacy-Preserving Federated Learning via Distributed Differential Privacy and Secure Aggregation' (arXiv:2604.07125)

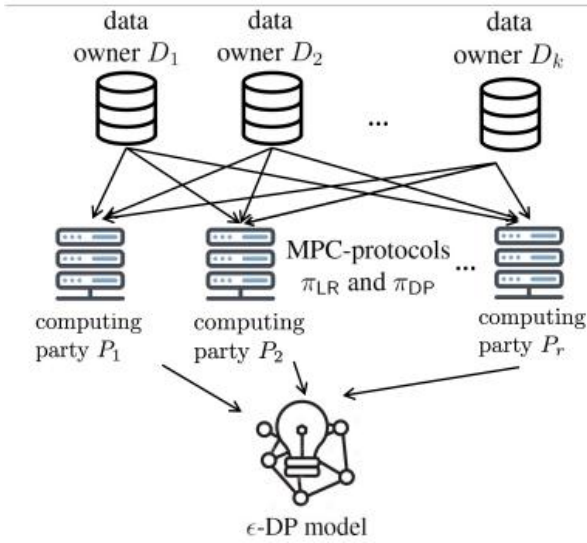


Figure 45: Chart from 'Training Differentially Private Models with Secure Multiparty Computation' (arXiv:2202.02625)

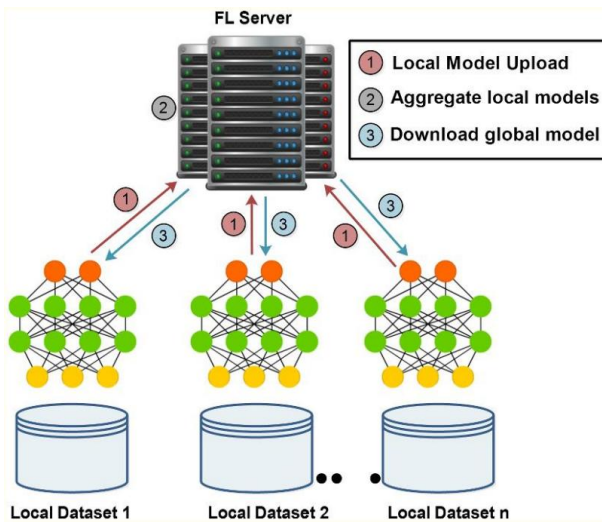


Figure 46: Chart from 'Privacy in Federated Learning' (arXiv:2408.08904)

trades utility for security, or employing cryptographic enhancements to enforce obliviousness. Although decentralized architectures eliminate the need for a trusted central server, they introduce complex trust dynamics where privacy guarantees weaken gracefully rather than collapsing abruptly as the number of colluding parties increases, necessitating careful calibration of security parameters against potential coalition sizes.

5 Future Directions

Despite significant advancements in integrating Differential Privacy with Federated Learning, several critical limitations persist that hinder widespread deployment in real-world scenarios. A primary challenge remains the inherent tension between rigorous privacy guarantees and model utility, particularly in highly heterogeneous data environments where static clipping thresholds and uniform noise injection often lead to biased convergence or excessive accuracy degradation. Furthermore, current cryptographic masking protocols struggle to seamlessly accommodate continuous noise distributions required for tight privacy bounds, creating computational bottlenecks and incompatibilities with discrete arithmetic systems. Existing frameworks also frequently lack robustness against adaptive adversarial attacks that exploit the specific structures of privacy mechanisms, and they often fail to account for the dynamic nature of client participation, where frequent dropouts can compromise both the integrity of secure aggregation and the validity of privacy accounting.

Future research must prioritize the development of fully adaptive, context-aware privacy mechanisms that dynamically calibrate noise scales and clipping bounds based on real-time gradient distributions and client heterogeneity. Investigating hybrid trust models, such as enhanced shuffle protocols combined with secure multi-party computation, could offer a pragmatic middle ground that reduces the reliance on trusted servers while minimizing the heavy noise penalties associated with purely local approaches. Additionally, there is a pressing need to design communication-efficient protocols that natively integrate quantization and sparsification with differential privacy, ensuring that compression techniques do not inadvertently weaken privacy guarantees or introduce new vulnerabilities. Further exploration into zero-sum per-

turbation strategies and correlated noise structures could significantly reduce cumulative noise magnitude in high-dimensional settings, thereby improving convergence rates without sacrificing formal privacy bounds.

Addressing these challenges will fundamentally transform the landscape of distributed artificial intelligence by enabling scalable, trustworthy collaborative learning systems. Successfully bridging the gap between theoretical privacy guarantees and practical implementation constraints will allow sensitive domains, such as healthcare and finance, to leverage collective intelligence without compromising individual data sovereignty. By establishing robust defenses against both passive inference and active poisoning attacks, next-generation frameworks will foster greater confidence in federated ecosystems, ultimately accelerating the adoption of privacy-preserving technologies in resource-constrained and adversarial environments. This evolution will not only enhance model performance but also set a new standard for ethical and secure machine learning practices globally.

6 Conclusion

This survey has provided a comprehensive synthesis of the mechanisms and strategies essential for integrating Differential Privacy within Federated Learning architectures. The analysis highlighted that effective privacy preservation extends beyond simple data decentralization, requiring a nuanced balance between rigorous mathematical guarantees and practical model utility. Key findings underscore the limitations of static gradient clipping, advocating instead for adaptive calibration techniques that dynamically adjust to gradient distributions to minimize bias and noise variance. Furthermore, the review differentiated between Local and Central Differential Privacy models, demonstrating how hybrid approaches like the shuffle model can leverage privacy amplification through subsampling and anonymization to tighten bounds without proportionally increasing noise. The examination of advanced aggregation protocols revealed that cryptographic masking, when combined with discrete noise mechanisms, offers robust protection against honest-but-curious servers, while quantization and sparsification serve dual roles in enhancing communication efficiency and inherently amplifying privacy. Finally, the integra-

tion of these mechanisms was shown to provide a foundational defense against both passive inference attacks and active data poisoning, establishing a holistic framework for resilient distributed intelligence.

The significance of this work lies in its systematic unification of disparate research threads that have historically been treated in isolation. By constructing a detailed taxonomy of differential privacy mechanisms tailored specifically for federated environments, this survey bridges the critical gap between theoretical privacy bounds and implementation realities. It moves beyond superficial treatments of noise injection to offer a critical comparative analysis of adaptive clipping, secure aggregation, and compression-aware privacy, elucidating the theoretical underpinnings and practical limitations of each. This consolidated perspective is vital for researchers and practitioners aiming to deploy collaborative machine learning systems that are not only compliant with stringent privacy regulations but also capable of maintaining high accuracy in heterogeneous network settings. By synthesizing insights from mechanism design, communication efficiency, and adversarial robustness, this review establishes a definitive roadmap for understanding the complex interplay between privacy preservation and model integrity, thereby facilitating the development of next-generation secure AI systems.

Looking forward, the field must address persistent challenges related to scalability, dynamic heterogeneity, and the evolving landscape of adversarial threats. Future research should prioritize the development of lightweight, adaptive protocols that can automatically tune privacy parameters in real-time based on network conditions and data distributions, reducing the reliance on manual hyperparameter optimization. Additionally, there is an urgent need to explore the intersection of differential privacy with emerging cryptographic primitives to further reduce the computational overhead of secure aggregation without compromising security guarantees. As federated learning expands into critical domains such as healthcare and finance, the community must foster interdisciplinary collaboration to standardize evaluation metrics and benchmarking frameworks that accurately reflect real-world privacy-utility trade-offs. It is imperative that subsequent innovations focus on creating resilient, scalable, and user-centric privacy frameworks that can withstand sophisticated

attacks while enabling the full potential of collaborative intelligence. Only through such concerted efforts can we realize a future where data collaboration thrives without compromising individual privacy.

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